Bash Scripting Lab Practical 3

1. Age Comparison

```
#!/bin/bash

echo "enter age"
read age

if [ $age -le 18 ]
then
        echo "you are child"
elif [ $age -le 45 ]
then
        echo "you are adult"

else
        echo "you are old"
fi
```

```
"age.sh" 15L, 163B                                                    11,6-13
```

Output

```
astha@cdac:~$ vim age.sh
astha@cdac:~$ ./age.sh
enter age
22
you are adult
astha@cdac:~$ ./age.sh
enter age
16
you are child
astha@cdac:~$ ./age.sh
enter age
45
you are adult
astha@cdac:~$
```

2. File creation

```
astha@cdac: ~
#!/bin/bash

for i in {1..50}
do
    sudo touch "file$i.txt"
done

~
~
~
```

Output

```
astha@cdac:~$ vim file
file10.txt  file18.txt  file25.txt  file32.txt  file3.txt   file47.txt  file8.txt
file11.txt  file19.txt  file26.txt  file33.txt  file40.txt  file48.txt  file9.txt
file12.txt  file1.txt   file27.txt  file34.txt  file41.txt  file49.txt  file_print.sh
file13.txt  file20.txt  file28.txt  file35.txt  file42.txt  file4.txt   file.sh
file14.txt  file21.txt  file29.txt  file36.txt  file43.txt  file50.txt  file.txt
file15.txt  file22.txt  file2.txt   file37.txt  file44.txt  file5.txt
file16.txt  file23.txt  file30.txt  file38.txt  file45.txt  file6.txt
file17.txt  file24.txt  file31.txt  file39.txt  file46.txt  file7.txt
```

3. Find Largest Number

```
#!/bin/bash

echo "Enter first no"
read a
echo "Enter second no"
read b
echo "enter third no"
read c

if [ $a -gt $b ] && [ $a -gt $c ]
then
        echo "a is greater"
elif [ $b -gt $a ] && [ $b -gt $c ]
then
        echo "b is greater"

else
        echo " c is greater"
fi

~
```

Output

```
astha@cdac:~$ vim largest_no.sh
astha@cdac:~$ ./largest_no.sh
Enter first no
20
Enter second no
30
enter third no
40
 c is greater
astha@cdac:~$ ./largest_no.sh
Enter first no
50
Enter second no
80
enter third no
20
b is greater
astha@cdac:~$ ./largest_no.sh
Enter first no
90
Enter second no
60
enter third no
30
a is greater
astha@cdac:~$
```

4. Factorial

```bash
#!/bin/bash


echo "enter no"
read num
fact=1

for (( i=1; i<num; i++ ))
do
        fact=$((fact*i))
done
 echo "factorial=$fact"
```

Output

```
astha@cdac:~$ vim fact.sh
astha@cdac:~$ ./fact.sh
enter no
6
factorial=120
astha@cdac:~$
```

5. Calculator

```
#! /bin/bash
echo "Enter First no"
read a

echo "enter second no"
read b

echo -e "enter operation\n 1.Addition \n 2.subtraction\n 3.multiplication\n 4.division "
read op

if [ $op -eq 1 ]
then
        echo "addition  $((a+b))"
elif [ $op -eq 2 ]
then
        echo "subtraction  $((a-b))"
elif [ $op -eq 3 ]
then
        echo "multiplication $((a*b))"
elif [ $op -eq 4 ]
then
        echo "division $((a/b))"
else
        echo  "invalid operator or invalid no"
fi
~
~
~
~
```

Output

```
astha@cdac:~$ vim fact.sh
astha@cdac:~$ ./fact.sh
enter no
6
factorial=120
astha@cdac:~$ vim calculator.sh
astha@cdac:~$ ./calculator.sh
Enter First no
6
enter second no
2
enter operation
 1.Addition
 2.subtraction
 3.multiplication
 4.division
1
addition  8
astha@cdac:~$ ./calculator.sh
Enter First no
6
enter second no
2
enter operation
 1.Addition
 2.subtraction
 3.multiplication
 4.division
2
subtraction  4
```

```
astha@cdac:~$ ./calculator.sh
Enter First no
6
enter second no
2
enter operation
 1.Addition
 2.subtraction
 3.multiplication
 4.division
3
multiplication 12
astha@cdac:~$ ./calculator.sh
Enter First no
6
enter second no
2
enter operation
 1.Addition
 2.subtraction
 3.multiplication
 4.division
4
division 3
astha@cdac:~$
```